Adaptive Virtual Reality Exposure Therapy for Acrophobia and Claustrophobia

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***Abstract*—Virtual Reality Exposure Therapy (VRET) has become an increasingly promising complement to traditional cognitive behavioral therapy for treating specific phobias. This project designed and implemented as VR system provides immersive exposure scenarios for two common phobias: acrophobia (fear of heights) and claustrophobia (fear of enclosed spaces). The system was built in Unity for the Meta Quest headset and features a height exposure environment with adjustable elevation as well as a claustrophobic corridor whose width dynamically decreases as the user walks forward. A simple VR menu enables users to select between the two therapy modes, and controller based UI interactions were implemented to make the experience intuitive and easy to navigate. The broader project vision based on my earlier proposal includes integrating heart-rate feedback and therapist dashboards for adaptive therapy in the future. The final system demonstrates how VR can simulate controlled, adjustable exposure experiences while maintaining user safety and providing flexibility that real world environments cannot.**

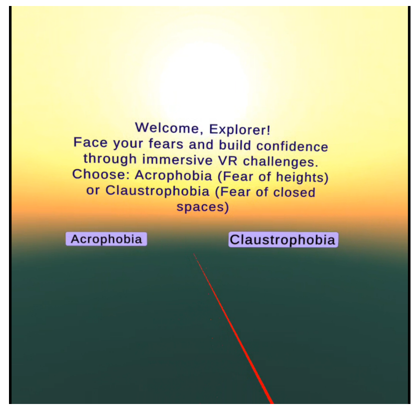


Fig. 1. Main menu interface of the VR Exposure Therapy system

I. Introduction

Phobias such as acrophobia and claustrophobia affect millions of people and often interfere with work, mobility, health procedures, and daily life. Traditional exposure therapy, while highly effective, is difficult to carry out consistently because real world environments are unpredictable, logistically difficult, and in some cases unsafe. Moreover, many patients drop out of therapy after learning that they will have to face the feared situation directly.

Virtual Reality Exposure Therapy (VRET) offers a practical alternative. VR allows therapists and researchers to recreate phobic environments in a safe, repeatable, and customizable way, without depending on access to actual high places or confined spaces. Multiple studies have validated the therapeutic value of VR exposure, showing reductions in anxiety symptoms comparable to or better than in vivo exposure [1], [2].

II. Related Work

A substantial body of research highlights VR's promise in treating phobias. For acrophobia, VRET has been shown to safely simulate height scenarios that would otherwise be logistically difficult or risky [2]. Systems like SAFEvR MentalVeRse recreate height and claustrophobic environments with strong usability scores and positive therapeutic effects [3].

Claustrophobia has also been successfully addressed through VR, El-Qirem et al. reported significant reductions in anxiety and physiological symptoms in enclosed-space VR simulations [1]. Similar progress has been made in treating other phobias, such as blood injection injury fear and spider phobia, through VR and augmented reality interventions [4].

Recent trends emphasize incorporating physiological feedback such as heart rate or skin conductance to adapt exposure intensity dynamically. Gutierrez-Sanchez et al. demonstrated that adaptive environments can increase personal relevance and improve user engagement [5]. These findings strongly support the future direction of this project.

III. System Overview

The VR Exposure Therapy system created for this project includes three main components:

1) A main menu scene with controller-based interaction.
2) A height-exposure environment where users can adjust elevation.
3) A claustrophobia corridor where the walls close in as the user walks.

The application was developed in Unity 2022 using the XR Interaction Toolkit for controller handling and interaction physics. All scenes were tested on a Meta Quest headset in both Play Mode and standalone build.

A. Main Menu and Interaction

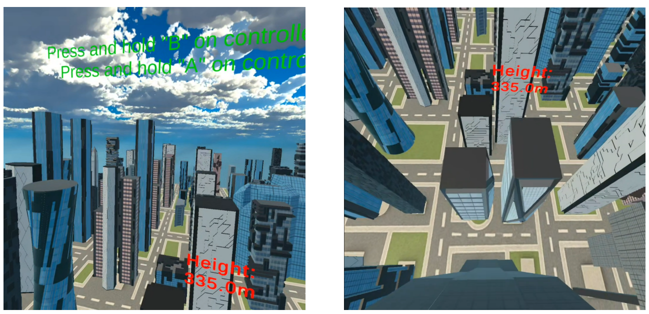
IV. Implementation Details

The system begins with a simple VR main menu that allows users to select either the acrophobia or claustrophobia exposure mode, as shown in Fig. 1. Implementing menu interaction in VR required configuring the canvas to World Space, enabling XRUIInputModule, and attaching colliders to UI buttons.By enabling the XR Ray Interactor controllers could point and click.

Although this seems like a small detail, smooth and predictable UI interaction is essential for VR therapy, especially for first time users who may already be anxious.

A. Height Exposure Scene

The height scene simulates standing on a rooftop surrounded by tall buildings as shown in the Fig. 2. Users increase their elevation by holding an input button, and a height meter updates in real time. To improve realism, the skybox and lighting were adjusted to create a bright daytime atmosphere.



(a) (b)

Fig. 2. Views of the Height Exposure Scene in the VR system.

Two technical issues had to be resolved:

- **Limited view distance:** Buildings disappeared beyond a certain distance due to default camera settings. Increasing the camera's far-clip plane and reducing fog density fixed this.
- **Building alignment and shadows:** Incorrectly placed buildings broke immersion. These were repositioned and rebuilt for consistency.

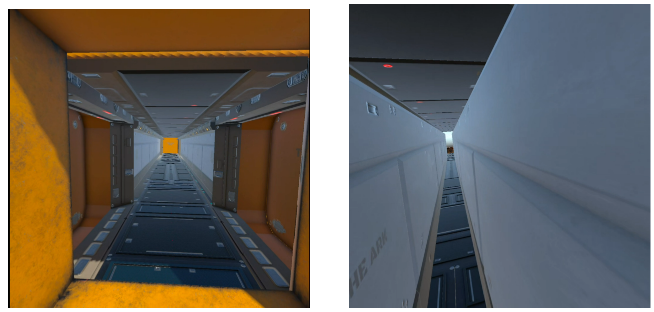
The elevation control reflects real therapeutic principles, where exposure is increased gradually rather than all at once.

B. Claustrophobia Corridor

The claustrophobia scene consists of a long sci-fi corridor that becomes narrower as the user progresses as shown in Fig. 3. The narrowing logic is tied to player movement:

$$W(t) = W_0 - k \cdot d(t)$$

where $d(t)$ is the player's forward distance and k controls how rapidly the corridor closes in. Users can feel the increasing confinement as the walls move inward.



(a) (b)

Fig. 3. Figure (a) shows the overall layout of the corridor, whereas Figure (b) depicts the user's experience inside the passage as the walls progressively close in.

The biggest technical challenge was the "disappearing floor" problem. Parts of the hallway vanished during movement because of mesh-bounds issues caused by scaling objects at runtime. The fix involved adjusting object hierarchies and enabling a setting that forces meshes to update even when outside the camera's frustum. This ensured consistency during the narrowing animation.

The corridor's pacing and lighting were also tuned to feel tense but not overwhelming, balancing user discomfort with therapeutic exposure.

V. Results

The system now includes two fully functional, immersive therapy scenes with smooth transitions from the main menu. Key outcomes include:

- A stable height exposure scene with adjustable elevation.
- A claustrophobia corridor with responsive inward motion.
- Reliable controller based menu and scene navigation.
- Improvements to lighting, visibility, and scene realism.

Figures included in the final submission will illustrate screenshots of all three scenes.

VI. Discussion

Developing the system highlighted how small implementation details such as UI collider settings, mesh bounds, and lighting can have a huge impact on VR usability. It also reinforced why VR is so suitable for exposure therapy it allows for precise control, immediate adjustments, and the ability to simulate environments that would otherwise be inaccessible or unsafe.

The current system is a strong starting point, but it does not yet include adaptive features or therapist control.These components are the part for future development.

VII. Future Work

The next steps that guided this project

- **Heart-rate adaptive exposure:** Incorporate Apple Watch or similar sensors to detect elevated anxiety (e.g., 20–30% above baseline) and automatically modify height, corridor width, lighting, or exposure pacing.

- **Therapist dashboard:** A web-based dashboard that is WebSocket based panel that receives JSON state from Unity could display real time height, corridor width, heart rate, and session time. Therapists could use this interface to adjust exposure levels or pause sessions.
- **Additional biometrics:** Skin conductance or breathing rate could offer a more holistic understanding of user stress.
- **Expanded environments:** Elevators, bridges, MRI rooms, and underground tunnels could broaden clinical relevance.

VIII. Conclusion

This project demonstrates a complete VR prototype for exposure therapy targeting acrophobia and claustrophobia. Although not adaptive yet, it establishes the foundation needed for real time physiological integration and therapist supervision. By combining immersive environments with adjustable difficulty and intuitive VR interaction, the system shows how VR can make exposure therapy more flexible, safe, and accessible. Future iterations will build on this foundation to create a truly adaptive, clinician-friendly therapeutic tool.

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